

Partitioning a 100" Domed Cylinder in Abaqus/CAE

Part: 3D deformable **solid**. ~100" dia. cylinder wall (~3.5" thick, 30" tall) with an integral dome that meets the bottom edge of the wall and rises to the top edge. The dome face carries multiple 7"-dia. holes arranged on circular (bolt-circle) patterns at several diameters.

Goal: (1) cut the part in half on the **XY plane** for symmetry, then (2) create **revolved (concentric-ring) partitions** on the dome about the revolution axis so the hole pattern can be isolated and meshed cleanly.

Key fact before you start: Abaqus/CAE has **no dedicated "revolve partition" button**. A revolved partition is produced by sketching a *section* on a datum plane that contains the axis, then **sweeping/revolving those sketch edges about a datum axis** using **Partition Cell → Extrude/Sweep Edges**. That is the tool that does what "shell revolve partition via a section sketch about a datum axis" describes. This guide uses it.

0. Conventions and assumptions

- The **revolution axis** is the vertical centerline of the cylinder. This guide assumes it is the global **Y-axis** (height direction), with the dome symmetric about it. If your axis is Z, substitute Z everywhere Y appears.
- Because the axis is Y, both the **XY** and the **YZ** principal planes contain the axis. You said to cut on **XY**, so the XY plane is your symmetry plane and it slices the part into two halves.
- Do all of this in the **Part** module (Tools → Partition), *before* meshing. Partitions made here carry into Assembly/Mesh.

- Turn on **View** → **Part Display Options** → **Datum** so datum planes/axes stay visible while you work.
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1. Create the datums you will need

You need one **datum axis** (the revolution centerline) and at least one **datum plane** through that axis (your XY cut plane). The principal planes/axes usually already exist at the origin, but create explicit datums so selections are unambiguous.

1a. Datum axis (revolution centerline)

1. **Tools** → **Datum**.
2. **Type: Axis, Method: Principal axis**, pick **Y-Axis** → **Apply**. *(If your model's centerline is not on a principal axis, use Method: 2 points and pick the two vertices on the true centerline, e.g. the dome apex point and the base-center point.)*
3. This datum axis is the axis you will revolve/sweep about in Step 3.

1b. Datum plane for the XY symmetry cut

1. **Tools** → **Datum** → **Type: Plane** → **Method: Principal plane**.
2. Pick **XY-Plane, Offset = 0** → **OK**. *(Use Method: 3 points or Point & normal if your symmetry plane is not a principal plane.)*

You now have a datum axis (centerline) and a datum plane (XY) coincident with the symmetry plane.

2. Cut the part in half on the XY plane (symmetry)

This splits the solid into two half-cells so you can keep one half for the symmetric model and so the flat cut face becomes the "canvas" for the section sketch in Step 3.

Menu: Tools → Partition → **Type: Cell, Method: Use datum plane.**

1. Start **Tools** → **Partition**.
2. **Type = Cell.**
3. **Method = Use datum plane.**
4. In the viewport, **select the whole solid cell** (the part), confirm the selection (click **Done** / middle-click).
5. **Select the XY datum plane** you made in Step 1b.
6. Click **Create Partition** (middle-click to confirm).

The part is now two cells split on XY.

Alternative if you didn't make a datum plane: Method = **Define cutting plane** → **Point & normal**: pick the origin (or any point on the axis) as the point and the global **Z-axis** as the normal — that defines the XY plane directly.

Keeping only half (optional, for a true half model): partitioning alone does *not* delete material; it only subdivides. If you want a physical half-part, either (a) apply symmetry BCs later and mesh both halves, or (b) go to the base feature and use **Shape** → **Cut** → **Revolve/Extrude** to remove one half, or model the part as 180° from the start. Most analysts keep the full geometry partitioned and simply apply **YSYMM/ZSYMM** boundary conditions on the cut face.

3. Revolved (concentric-ring) partitions on the dome — the "section sketch about a datum axis" step

The idea: on the **XY cut face** (a meridian cross-section of the dome that now contains the axis), sketch lines at the **radial stations** of each hole circle. Then **revolve those edges about the datum axis** so each

becomes a full circular ring partition on the dome face. This gives you clean concentric bands that isolate each bolt-circle of holes.

3a. Sketch the section (radial ring stations) on the cut face

Menu: Tools → Partition → **Type: Cell, Method: Sketch planar partition** (a.k.a. "sketch the section").

1. **Tools → Partition → Type = Cell → Method = Sketch (planar).**
2. **Select the cell** to partition (the dome half) → **Done.**
3. When prompted for the sketch plane, **select the flat XY cut face** created in Step 2.
4. Pick an **edge or axis to appear vertical/at-right** when asked (pick the datum axis or a vertical edge) so the sketch orientation is intuitive.
5. In the Sketcher you are now looking at the meridian cross-section of the dome wall. **Draw a short line across the dome-wall thickness at the radial location of each hole ring** — i.e. at each bolt-circle radius R1, R2, R3... measured from the axis. Each line runs from the inner dome surface to the outer dome surface (or long enough to cross the material at that station).
 - Use **Add → Dimension** and constrain each line's distance from the axis to the exact ring radius (e.g. $R = \text{hole-circle diameter} / 2$).
 - Add a line **on each side** of a ring if you want a *band* that brackets the holes (e.g. $R - \text{offset}$ and $R + \text{offset}$, with $\text{offset} \approx \text{hole radius} + \text{a little}$), so the holes sit inside their own strip. This is what makes the later mesh structured around each hole.
6. **Done** to exit the Sketcher → **Create Partition.**

You now have partition **edges on the meridian face** at each hole-ring radius. These are the "section" you will revolve.

3b. Revolve the section edges about the datum axis

Menu: Tools → Partition → **Type: Cell, Method: Extrude/Sweep**

edges — this is Abaqus's revolve.

1. **Tools** → **Partition** → **Type = Cell** → **Method = Extrude/Sweep edges**.
2. **Select the edges** you just sketched on the meridian face (the radial ring lines) → **Done**.
3. When Abaqus asks how to sweep, choose the "**Sweep along an edge / circular path**" option and **select a circular edge that goes around the axis** — the dome's outer circular edge (a semicircle on the half model) is the natural pick. Sweeping along that semicircular edge **revolves the section edges 180° about the axis**, generating the concentric ring partitions.
 - If Abaqus offers "**Extrude along direction: select the datum axis**" — that translates (straight extrude), *not* revolve. For a **revolve** you must pick the **circular edge path**, not the axis-as-direction.
4. **Create Partition**.

Result: the dome solid now has clean circular ring partitions at every bolt-circle radius, each holes' pattern boxed into its own annular band.

If Extrude/Sweep won't revolve cleanly (common when the profile and path aren't co-planar/normal), use the fallback in Step 3c.

3c. Fallback / alternative revolve methods

- **Datum-plane fan for radial cuts:** create datum planes through the axis at each angular hole location (**Tools** → **Datum** → **Plane** → **Rotate from plane about the datum axis**, entering the pattern angle, e.g. 360/N). **Partition Cell** → Use datum plane on each. Combined with the ring bands from 3a-3b, this produces a checkerboard of quad-friendly patches with one hole per patch.
 - **Partition Face** → **Sketch** on the curved dome face: if you only need the *surface* split (for a shell-like mesh control), sketch the ring circles directly and let Abaqus intersect them with the face. Less robust on doubly-curved faces than the cell method above.
 - **Extend face:** because the dome is spherical, **Partition Cell** → **Extend face** can extend the spherical dome surface to cut the wall — handy for separating the dome cap from the cylinder wall.
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4. Isolate each hole for a structured/sweep mesh

With rings (3a-3b) and optional radial cuts (3c) in place:

1. Add short **radial partition edges between adjacent holes** in each band using **Partition Face** → **Sketch** or **Partition Face** → **Use shortest path between 2 points** (pick a vertex on one hole edge and the matching vertex on the neighbor). This walls off one hole per cell.
 2. Aim for each hole to sit in a small **washer/annulus of 4 (or 8) faces** around its rim — the classic "butterfly"/O-grid around a hole — so you can seed the hole edge and get well-shaped elements at the stress concentration.
 3. Keep partition cells **simple (6-face, hex-able)** wherever possible.
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5. Mesh controls after partitioning

1. **Mesh module** → **Assign Mesh Controls**: select the well-shaped cells → **Hex**, technique **Structured** or **Sweep**. Leave awkward transition cells as **Tet** if needed, and mate with hex via tie/shared nodes.
2. **Element type**: for a 3.5" thick wall carrying bending, prefer **C3D8I** (incompatible-mode hex) or **C3D20R** (quadratic reduced) — at least **2–3 elements through the thickness**.
3. **Seed the hole edges** (Seed → Edges) with enough elements (≥ 8 –12 around each 7" hole) to resolve the stress concentration; bias seeds finer toward hole rims.
4. **Verify**: Mesh → Verify (Analysis checks) — watch aspect ratio, min angle, and Jacobian around the holes. Re-partition if any cell won't mesh Hex.
5. **Symmetry BC reminder**: on the XY cut face apply the correct symmetry (e.g. **ZSYMM** if the normal to the cut face is Z), so the

half model behaves like the full dome.

6. Quick command summary

Step	Menu path	Purpose
Datum axis	Tools → Datum → Axis → Principal axis (Y)	Revolution centerline
Datum plane	Tools → Datum → Plane → Principal plane (XY)	Symmetry cut plane
Half cut	Tools → Partition → Cell → Use datum plane (XY)	Split part in half
Section sketch	Tools → Partition → Cell → Sketch planar on the XY cut face	Radial ring stations
Revolve	Tools → Partition → Cell → Extrude/Sweep edges , sweep along circular edge about the datum axis	Concentric ring partitions
Radial cuts	Tools → Datum → Plane → Rotate about axis , then Partition Cell → Use datum plane	One hole per patch
Hole O-grid	Partition Face → Sketch / shortest path	Structured mesh at holes

7. Common pitfalls

- **"Extrude/Sweep" translated instead of revolved:** you picked the axis as a *direction* (straight extrude). Pick a **circular edge** as the sweep **path** to get a revolve.
- **Sketch face not flat:** the section sketch must be placed on the **flat XY cut face** from Step 2, not on the curved dome face.

- **Partition ≠ material removal:** the half cut only subdivides; keep both halves + symmetry BC, or remove one half at the base feature if you truly want a half part.
 - **Non-hex-able cells:** if a cell refuses Structured/Sweep, it usually has >6 faces or a concave loop — add one more partition to simplify it, or set it Tet.
 - **Datum axis picks the wrong line:** if the centerline isn't a principal axis, define the datum axis by the two centerline vertices (apex + base center) instead of "Principal axis."
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Sources

- [Methods for partitioning cells — Abaqus/CAE Reference \(MIT mirror\)](#)
- [Using the sketch planar partition method to partition a cell — Abaqus/CAE](#)
- [Geometry Partitioning in Abaqus/CAE — TECHNIA](#)
- [Abaqus-CAE: Radial Partition — Eng-Tips forum](#)